# 4

# ENGINEERS IN SOCIETY

## 4.1 Social Ascent and Images of Engineers

Engineers pride themselves on solving real-world problems and getting things done creatively, scientifically, effectively. Practicality and efficiency, however, are not universally valued. "The idea of utility has long borne the stamp of vulgarity," a historian remarked on the literary culture, where the concept of technology failed to gain currency until after World War I and even now is deemed a "hazardous concept."[1] Differences between the cultures of doers and talkers spawn misunderstanding. Fortunately, stereotypes of engineers have not eclipsed their socioeconomic contributions.

Historically, engineers and their predecessors came mostly from working families, toiled with their hands, relied more on their own thinking and experience than on schooling, and were obliged to deliver products on demand. They were looked down on by ladies and gentlemen of letters and leisure who fancied themselves cultured and superior to those who produced the goods and services that made possible their elegance and opulence. Elitist slings and arrows were felt by Leonardo da Vinci, who replied: "I am fully conscious that, not being a literary man, certain presumptuous persons will think that they may reasonably blame me; alleging that I am not a man of letters. Foolish folks! do they not know that I might retort as Marius did to the Roman Patri-



cians by saying: That they, who deck themselves out in the labours of others will not allow me my own."[2]

Before he entered government, Herbert Hoover visited England and joined a debate to argue that Oxford and Cambridge universities should include engineering in their instruction. On the Atlantic passage back to America he shared a dining table with an English lady and they had many delightful conversations. At the farewell breakfast she asked his profession. Upon hearing that he was an engineer, she emitted an involuntary exclamation: "Why, I thought you were a gentleman!"[3]

Engineers surprised not only English ladies. Theodore Roosevelt, U.S. president, wrote to a friend about John Stevens, whom he appointed as the chief engineer to build the Panama canal. That engineer, he discovered, read a lot of literary books. And he read not only novels but also poetry. "This I regard as astounding," Roosevelt wrote.[4]

### *Triumph of Self-Education*

Stevens came from the backwoods and educated himself while working on the railway. This was almost the norm for early engineers. Most technical pioneers of the industrial revolution came from the lower social classes and had little schooling. Even among leading engineers, only a few had any higher education. Some had attended grammar school or high school. Others had spent their childhood working in mines or at arsenals. The majority had been apprenticed to millwrights, barbers, carpenters, stone masons, instrument makers. Because of their humble origins, these pioneering engineers were often stereotyped as craftsmen incapable of scientific reasoning. That stereotype persists. As late as 1959, when engineering research was widespread, a scholar declared that no word existed for an improver of technology comparable to "scientist" as an improver of knowledge. He rejected "engineer" because "it originally, and essentially still, refers to the expert practitioner of one of the arts."[5]

That stereotype is wrong even for early pioneers. We saw the scientific thinking of Smeaton and Watt, in section 2.3. Painstaking historical research has shown that although the leading engineers had not received formal education, they were not uneducated. As historians A. E. Musson and Eric Robinson remarked, the erroneous stereotype of early engineers as illiterate and unscientific tinkers is "a direct outcome of

the difficulty which academically educated people often have in understanding the possibilities of self-education."[6]

Educate themselves pioneering engineers did. In Britain young Fairbairn burned the midnight oil to learn mathematics, Smeaton limited his business to find time for scientific studies, and George Stephenson had his son Robert borrow technical books from the library and read them aloud to him in the evening. Telford wrote in a letter: "Knowledge is my most ardent pursuit," and despite his preference for the experimental approach, he read widely and collected a huge library that included many French engineering and theoretical books.[7] Several engineers, including Smeaton, Maudslay, Fairbairn, and Nasmyth, were interested in astronomy. They made their own telescopes and observation instruments, thus demonstrating not only the requisite mathematical competence but also a broad intellectual curiosity. The spirit of self-help leaps from the pages of Samuel Smile's *Life of Engineers*. With little if any formal training, they transformed themselves from artisans to engineers and sometimes successful entrepreneurs. Often one man excelled in both civil and mechanical engineering. The Stephensons built not only steam engines and locomotives but also the railways and the bridges that they ran on. Smeaton not only built the Eddystone Lighthouse but also performed systematic experiments to improve efficiencies of water wheels and reported his results in scientific papers. Examples multiply, and they are not limited to the British. American engineers such as Edison readily come to mind. With their independence of thought and keen intuition, these autodidacts absorbed the relevant scientific knowledge and integrated it with the practical know-how acquired through their work, thus elaborating it and creating a new body of engineering knowledge. As historian Sidney Pollard remarked, "They rose from the ranks, so that in one lifetime they had not only to make the transition from rags to riches, but also to absorb, and indeed create, a completely new technology."[8]

### *Efficacy of Informal Association*

Self-education requires motivation and access to existing knowledge. It is unclear why in that particular time and society so many people were so motivated, but opportunities were present in the British economic expansion. In response to demand, the eighteenth century saw the publication of a profuse number of textbooks and periodicals, and other printed matter on mathematical, scientific, and technical topics addressed to the popular audience. Lectures disseminating scientific knowledge, popularized in London, were later spread around the country by itinerant lecturers. The fascination they exerted on ordinary people was captured by Joseph Wright in his paintings, *A Philosopher Giving a Lecture on the Orrery* and *An Experiment on a Bird in the Air Pump*.[9]



To working men with high motivation and ability, the doors of the scientific community were not closed. Associations were not formal but, contrary to stereotypes, they were not in vain. Many conference goers think they reap the most benefit not from formal talks but from informal chats in the corridors. A formal theory has its own systematic structures, most of which may be irrelevant to the problem at hand, and learning them may be a distraction and a waste of intellectual resources. In informal conversation between two knowledgeable persons, one can quickly home in on relevant pieces of information. Broad visions, open-mindedness, critical analysis, and other rational ways of thinking crucial to science are also more likely to be picked up through personal contacts than in formal theories. Such informal interactions occurred during the industrial revolution. Many clubs and associations existed in which philosophers, scientists, engineers, and industrialists mingled to discuss their mutual interests. Among them was the Lunar Society of Birmingham, which met monthly when the moon was full throughout the late years of the eighteenth century and into the nineteenth. Its members at one time or another included Smeaton; Watt and his partner Matthew Boulton; William Herschel the astronomer; Joseph Priestley the chemist who isolated oxygen; Josiah Wedgwood, an industrialist steeped in chemistry; and Erasmus Darwin, a physician and the grandfather of Charles. They brought visitors, demonstrated experiments, asked questions, and gave advice, thus providing mutual support as they pursued their manifold interests. Such was the scientific ground in which the seeds of technological ideas sprouted and grew.

The prevalence and overall importance of interaction between scientists and engineers should not be overstated. It nurtured leaders, but a long road still lay between craft and scientific engineering. Even knowledge that would have been readily applicable and profitable in mechan-

ical crafts, such as the optimal shapes for teeth of gears as worked out by mathematicians, waited in oblivion for decades. Nevertheless, some interaction did occur. The ability to discuss issues as equals with leading scientists showed that many engineers were anything but incapable of scientific thinking.

### *Establishment of Professional Associations*

More important than mixing with scientists, engineers formed associations among themselves to exchange information and discuss problems peculiar to their concerns. Smeaton initiated the Society of Civil Engineers in 1771, which was little more than a dining club at first. The British Institution of Civil Engineers (ICE), established in 1818, was the first professional society in the modern sense. Its president, Telford, exhorted members to volunteer their effort to keep it going. Unlike their French colleagues, they had no government sponsorship, but then they were free from government interference. From their craft origins, autodidactic British engineers rose to enunciate the vision of a new profession in the charter of the world's first engineering society: "Engineering is the art of directing the great sources of power in Nature to the use and convenience of man; being that practical application of the most important principles of natural philosophy which has, in a considerable degree, realized the anticipations of Bacon, and changed the aspect and state of affairs in the whole world."[10]

The fledging ICE acquired its own premises and library, organized weekly meetings, and recorded the proceedings. Soon it started to publish engineering journals, coordinate the dissemination of technical knowledge, establish rules for membership, and standardize certain measures and practices. It served as the model for other societies. Unhappy about ICE's overemphasis on construction, mechanical engineers established the Institution of Mechanical Engineers in 1846 with George Stephenson as the first president. Engineers in other areas followed suit.

The German Society of Engineers, formed in 1856, played a crucial role in bringing German polytechnic schools first up to university status and then into pioneering graduate education. The American Society of Civil Engineers appeared in 1852. Societies proliferated because of the diversity of engineering specialties. Efforts to form a national fraternity



of engineers started in 1886, culminating in the foundation of the U.S. National Academy of Engineering in 1964.

### *Rise of Engineering Universities*

Professional societies organize active workers, and their journal articles are mainly addressed to knowledgeable readers. As with any human enterprise, a profession must nurture its future generations. A knowledge-intensive profession such as engineering further requires the organization and presentation of knowledge in forms accessible and stimulating to the novice. Self-education and informal learning can go a long way, but they are too unpredictable to produce a sufficient number of qualified engineers, especially when technological complexity gallops ahead. Formal education connects generations of workers more consistently. In this Britain lagged. Its major universities, engrossed in classics and the liberal arts, provided technical education belatedly and reluctantly. As the industrial revolution progressed, it began to lose its technological leadership. An engineer said at an 1893 ICE meeting: "The danger which, as many thought, now threatened English engineering, lay in the more thorough education and superior mathematical knowledge of so many foreign engineers."[11] The foundation of that threat, laid centuries earlier, revealed the role of government as a partner in technology.[12]

Practical artists who undertook public works and military projects had a closer relationship with governments than workers in many other trades. Recognizing their importance and wanting to protect them from the arrogance of soldiers, the French government created the Corps des Ingénieurs du Génie Militaire in 1675. Like today's U.S. Army Corps of Engineers, French officers in the corps often engaged in constructing roads, bridges, and other civilian works. It was at this time that "engineer" first became a title for professionals with special expertise. To educate them better, the government established around 1747 several military academies equipped with first-rate physics and chemistry laboratories. These schools offered rigorous curricula that included cartography, machine design, building construction, and especially mathematics. One of their graduates, Coulomb, became a renowned engineer and then physicist.

The greatest change occurred when the French Revolution un-

leashed enthusiasm to improve the human condition by rational means. The École Polytechnique, created in 1794, was an educational breakthrough not only in engineering but in science generally. Some four hundred students, selected by competitive examination from all over France, enrolled in the first year. They started with two years of scientific instruction that included calculus, mechanics, and chemistry, so that they had a grasp of general principles before they specialized in various branches of engineering. Among their professors were the best engineers, mathematicians, and natural scientists, and many of the students embarked on equally brilliant careers. Graduates from the Polytechnique secured the reputation of scientific engineers. Napoleon would not let them into battle even in dire crises, saying he would not wish to kill the goose that lays golden eggs.

Sponsored by the government, the École Polytechnique was subject to the vicissitudes of politics, which were especially violent in France at that time. It was also criticized for being too theoretical. Several private technical colleges appeared, offering more readily useful knowledge to a greater number of people. Chief among these was the École Centrale des Arts et Manufactures, which boasted Eiffel and Jenny as alumni.

The French model was emulated worldwide. In America, the military academy at West Point, first established in 1802, was reorganized in 1812 with the help of French-educated officers who introduced an engineering curriculum along the line of that at the École Polytechnique. It produced far fewer engineers than its French counterpart but shared its vulnerability to competition from the private sector. Rensselaer Polytechnic Institute, patterned after the École Centrale, appeared in 1849 and became the model for several others. Among its first graduates was Washington Roebling, who succeeded his father, John, in building New York's Brooklyn Bridge. Demand for courses outpaced supply, and thus construction sites and industrial shops continued to be major spawning grounds for engineers. Technical education in the United States received a big boost from the Morrill Land Grant College Act of 1862, which granted federal lands to institutions of higher education with the stipulation that they offer instruction in "agricultural and the mechanic arts." Among others, it aided MIT, Rutgers, Yale, and the University of Michigan, and it created Cornell, Purdue, and the state universities of Ohio and Iowa. Ten years later the nation had seventy engineering colleges.



Enrollment in engineering courses increased much faster than in liberal arts courses. To accommodate the torrent of students, in 1895 the University of Michigan created an independent engineering school, whose enrollment grew from 331 to 1,165 during the next decade while its school of literature added just over 300 students. Similar enrollment patterns were seen at other universities. Together, universities awarded 226 engineering degrees in 1880 and some 11,000 in 1930. And the trend continued. Engineering soon became a favorite subject of college students. (For more recent profiles, see Appendix A).

### *Social and Sociological Images of Engineers*

Besides leading technological revolutions, engineering institutes contributed to the democratization of education. By lightening up the prevailing requirements for arcane classics and replacing them with modern subjects useful in productive occupations, they opened a route to upward social mobility for children of the lower and middle classes who hitherto could not afford higher education. The social difference between the new students and old elites was noticed early on. MIT's president reported in 1894: "An unusually large proportion of our students come from families of moderate means [who] never consider the possibility of first going to a classical college and then to a professional school." Farther up the Charles River, Harvard's president observed marked differences between liberal arts students and those in engineering—"young men who know from the start the profession they are destined for."[13] The techies were coming. Talented, motivated, determined to contribute to and benefit from fast-moving industries that rewarded productive achievers, they quickly proved that workers dedicated to producing socially beneficial goods and services could excel not only with their hands but also with their brains. The old cultural aristocrats who scorned them were soon discomfited by the emergence of a rival culture.

Folks cheered the new professionals who lit up their homes with electricity, connected them by telephone, brought freedom of mobility with cars and bridges. In an extensive study of popular culture, histo-

rian Cecelia Tichi found that "the engineer became a vital American symbol between the 1890s and 1920s. In popular fiction, as we shall see, he signified stability in a changing world. He was technology's human face, providing reassurance that the world of gears and girdles combined rationality with humanity." In this period, the engineer appeared as the hero in more than a hundred silent movies and in best-selling novels that sold nearly five million copies. "In these popular novels the engineer's vision is both prophetic, in that it foresees the future, and poetic, since it originates within the realm of imagination . . . These engineers realize their dreams. They bring the bold, visionary schemes from the imagination into tangible reality. They are poets for the quotidian."[14]

The young profession received a totally different treatment from the old establishment. One historian cited an engineer who in the late 1890s predicted that because electricity would revolutionize society, engineers would be the "priests of the new epoch." If so, the historian asked, "why are engineers so invisible in American culture? Is it because, after all, they are not synonymous with technology? Or is it that, in any event, elevated literature is the wrong place to look for them?" This elitist reduction of culture to elevated literature I will discuss shortly. Such disdain was corroborated by Tichi, who observed that the engineer was "the invisible man in American studies." Doctors, lawyers, and other professionals have been discussed in college classrooms and scholarly journals, but not engineers; "critics and analysts have been oddly blind to their presence."[15]

When technological achievements could no longer be ignored, blanket contempt changed to a partial condescension expressed in two stereotypes, which have been extended to natural scientists when their research results have proved to be of practical use and thus less than "pure." The first is *Dr. Frankenstein,* who makes monsters. The second is the *nerd* or the *geek:* bright and proficient but bloodless and deficient in social relationships. The stereotypical transition from the illiterate artisan to the "antisocial brainiac" was described by a historian in discussing the term "institute of technology." He wrote: "This abstract word [technology], with its vivid blankness, its lack of a specific artifactual, tangible, sensuous referent, its aura of sanitized, bloodless cerebration and precision, helped to ease the introduction of the practical



arts—especially the new engineering profession—into the precincts of higher learning."[16] Compare this picture of technology to Aristotle's description of *téchnē* as the capacity to produce with a true *logos,* presented in section 2.1, and one sees how much the literary culture has changed.

Cultural resentment increased after World War II, when technical professionals became socially influential. Engineers were referred to as "narrow gauge" and "domesticated servants of capital."[17] Surveying the scholarly literature, another historian observed: "Ever since John Alexander Law Waddell started telling engineering students to clean their fingernails and comb their hair, it has been easy to caricature the profession . . . Instead of the portrait of a profession, what we have is a grab bag of stereotypical images and they picture a group that seems politically inflexible, socially awkward, culturally limited, and ethically inert."[18]

Are the uncouth stereotypes of engineers justified? We glance at some general rebuttals here; many concrete examples are presented in Chapter 7. William Rogers, founder of MIT, wrote in his proposal for the new institute: "In the features of the plan here sketched, it will be apparent that the education we seek to provide, although eminently practical in its aims, has no affinity with that instruction in mere empirical routine which has sometimes been vaunted as the proper education for those who are to engage in industries. We believe, on the contrary, that the most truly practical education, even in an industrial point of view, is one founded on a thorough knowledge of scientific laws and principles, and one which unites with habits of close observation and exact reasoning, a *large general cultivation.*"[19]

Ever since MIT opened its doors, it has required students to take history, literature, and other humanities courses besides science and engineering. This practice is common in engineering education. The advice that schools should make economics and the humanities required courses, issued by the Society for Promotion of Engineering Education in the 1920s after a multiyear study, had a profound influence. A 1955 report for the American Society for Engineering Education proposed that about a quarter of undergraduate courses be devoted to humanities and social sciences, and the remaining three quarters divided among the relevant natural sciences and engineering specialties.[20] This

mixture has become almost the norm in American universities, as the 1955 report formed the basis for professional accreditation of engineering curricula. Instruction in engineering usually does not start until the sophomore year. Freshman engineering majors compete head-on with history majors in required history classes, and they are not offered "English for engineers" classes akin to the "physics for poets" classes often offered for nontechnical students. Some universities, for example Stanford, demand more than the general university humanities requirements for engineering majors, who must take additional courses in the relationship between technology and society. Generally, engineering and natural science majors take more humanities classes than liberal arts majors do science and technology courses. On the Graduate Record Examination for entrance to graduate school, engineers and natural scientists consistently and comfortably beat social and behavioral scientists in *verbal* scores, not to mention analytic and quantitative skills.[21]

# 7

# LEADERS WHO ARE ENGINEERS

## 7.1 Business Leaders in the Car Industry

Their sweat evaporating under the desert sun, a team of construction managers and civil engineers worked with Egyptologists to study the Great Pyramid of Giza. They were concerned less with physical methods than organizational ones, such as the logistics of construction and the sustenance of workers. The Great Pyramid's mass is roughly two-thirds that of the Hoover Dam. Its construction took an estimated average labor force of 13,200 workers more than ten years. How did the Egyptians orchestrate and provide for so many people? The team asked, How would *we* do it?[1]

Contemporaries of ancient master builders remarked on their managerial abilities in reckoning costs, securing supplies, coordinating efforts, communicating specifications, and resolving personnel conflicts. The care with which Callicrates chose contractors in erecting the Long Walls in ancient Athens impressed the Roman historian Plutarch, who described it in detail. Vitruvius explained that to construct eaves and sewers properly, the master builder had to be familiar with the laws of the city. Dealing with business and the government has always been a part of the engineer's job.[2]

In today's market economy and representative democracy, engineers have to deal with business professionals and consumers, government



bureaucrats and interest groups. All these contacts require significant people skills. "An engineer has to know a lot about people, the ways they organize and work together, or against one another, the ways in which business makes a profit or fails to, especially about how new things become conceived, analyzed, developed, manufactured, put into use," wrote Bush. "The engineer stands in the middle of the spectrum between men and things, not only in his objectives, but also in the ways in which he uses science, and in the ways he operates in organization . . . In any engineering career there is likely to be a gradual shift from emphasis on knowledge of *things* to emphasis on knowledge of *men*."[3]

Engineering's human dimension, where engineers exercise their technical expertise in the roles of managers, policy makers, or simply responsible citizens, is no less significant than its physical dimension. Here purposive questions of what engineered systems are ultimately for, how well they serve their goals, and how to regulate their social impact come to the fore.

### *Another Stereotype Debunked*

"Well-round and engineering school? Until recently, it would have seemed laughable to link the two phrases." Thus stated an article in *U.S. News and World Report* in 2000. Referring to MIT and other institutes "reputed to turn out antisocial brainiacs," it reported their recent efforts in "building a new breed of engineer—one with management savvy."[4] The news magazine rightly spotted the recent trend in engineering schools of stepping up education in business leadership, partly in response to student demand. It also revealed the prevalence of the nerd stereotype; only those blinkered by it would regard management savvy in engineers as a laughable idea.

Paul Litchfield, an MIT alumnus, wrote after becoming president of Goodyear Tire and Rubber: "Harvard men used to say the Tech would turn out engineers but that the nation would turn to Harvard for the head of business. Some of the men who were in school with me, however, got along all right, and I got a little satisfaction out of it as Alfred P. Sloan, class of '95, became head of General Motors, Gerard Swope, also '95, head of General Electric, Irenée du Pont, class of '97, was made head of that great company, and Roger Bobson, class of '98, became a world-famous economist."[5] This is not news; Litchfield was re-

*Table 7.1* Percentage of top executives with an engineering background in large U.S. corporations.

| | 1900 | 1925 | 1950 | 1964[a] |
|---|---|---|---|---|
| Percentage of top executives: | | | | |
| Whose college degree was in engineering[b] | 24.0 | 32.8 | 32.2 | 44.1 |
| Whose first full-time job was as an engineer | 7.7 | 12.2 | 16.8 | 26.6 |
| Whose chief occupational experience was engineering | 12.5 | 15.6 | 19.3 | 34.8 |

*Source:* M. Newcomer, *The Big Business Executive* (New York: Columbia University Press, 1955), tables 27, 36, 38; Scientific American, *The Big Business Executive/1964*, tables 8, 14, 15.

a. 1964 numbers are for engineering or science.

b. Among those who held an undergraduate or graduate degree.

ferring to graduates of the 1890s. They were all educated as engineers. Business schools at that time were merely struggling experiments.[6]

The significant involvement and success of engineers in business, as revealed in the survey results of Table 7.1, would not surprise those who attend more to work content than sociological stereotypes. Practical and hard-headed, engineers mean business. From railroads to automobiles to electricity to computers, new technologies have spawned sunrise industries. Engineers with a mastery of those technologies have an edge in exploiting their commercial potential. Nor are they incompetent in areas of management and finance. Economy and efficiency, although snubbed as vulgar by cultural elites, have always been valued by common working people. Since the time of master builders, engineering projects have demanded skill in both weight and cost reckoning, both physical and organizational technologies. Coordination of work becomes more important as technology becomes more complicated. Effective organization is indispensable in hard core engineering design, such as airplane development projects that involve hundreds or thousands of engineers. From this type of management, business administration is only one step away. It has offered engineers an attractive route of advancement, especially in times before engineering science took off and began to compete for talent.

Technical experts hold their own on the corporate ladder in competition with business schools' mass-produced MBAs. Of the seven corporations whose stories were featured in *Fortune*'s 2002 listing of the 500 largest American companies, two were headed by engineers and one by a scientist.[7] And this is just the tip of the iceberg, compared with the engineers who make up senior and middle management. A 1999 study by the National Science Foundation found that about 15 percent of those with a master-level degree in engineering become senior managers at some point in their career. The number rises to 35 percent for engineers who also hold a business degree.[8] Similar patterns for engineers in business obtain in Germany and Japan.[9]



Engineers make not only run-of-the-mill chief executive officers but reach the pinnacle of excellence. The two leaders in consolidating the entire German chemical industry during the interwar years were Carl Bosch, chairman of BASF, and Carl Duisberg, director of Bayer, two of the three top chemical firms. Bosch was an engineer, Duisberg a chemist. Both ascended from a technical research staff position to chief executive, and Bosch got a Nobel Prize on the way.[10] American engineers have been no less successful. Of the four finalists for *Fortune*'s "businessman of the century"—Henry Ford of Ford Motors, Alfred Sloan of General Motors, Thomas Watson, Jr., of IBM, and Bill Gates of Microsoft—three had engineering backgrounds. They represented a radical shift from businessmen of the nineteenth century. In *Fortune*'s words: "Organization Man rose to challenge the robber baron." *Fortune* also chose as "the manager of the century" Jack Welch, who, with a chemical engineering doctorate, started as a researcher in the materials department of General Electric, climbed to chief executive, and in the last two decades of the twentieth century strongly influenced management thinking by reengineering that lethargic conglomerate into a powerhouse.[11]

*Fortune*'s two honors highlight two types of business talent: the founder of businesses and the professional manager, especially at the top level of large organizations. Ford, who was named businessman of the century, was an entrepreneurial founder, Welch a professional manager. Sloan, who first engineered the basic structures of giant multidivisional corporations, bridged the two.

Engineer-entrepreneurs are familiar today, since they have grabbed business headlines in the information revolution. They are only the

most recent generation in a long heritage that will continue. Back in the first industrial revolution, Watt commercialized his steam engine and Stephenson his locomotive, and Maudslay and Nasmyth started the British machine-tools industry. In the second industrial revolution, Ford designed and produced the Model T, and he was only one of many Detroit engineers whose names appear on popular cars. More recently, Intel's first two presidents in the twentieth century, Noyce and Moore, both experimented with silicon themselves, and many other leaders in the semiconductor industry "had silicon under their fingernails," as electrical engineer Ian Ross put it.[12] Business is always risky, and success is never guaranteed, whatever one's background. Nevertheless, an entrepreneur hides inside many engineers, who know that what they make has commercial potential and who, with their can-do spirit, are less deterred by business risks.

### *Strategic Management of Large Corporations*

At the end of the twentieth century, almost half of Americans owned stocks, either directly or through retirement plans. The democratization of ownership in enterprise capital requires skilled professional managers to run the corporations for diverse stockholders, most of whom are not involved in business. Guess who pioneered modern business administration? Combining their practical tradition with scientific reasoning, engineers were the first to be able to think big; think practically and objectively; think clearly, analytically, systematically. Inventing new approaches distinct from those of traditional merchants, they introduced strategic, operational, and tactical management in the large corporations that began to emerge in the mid-nineteenth century.[13]

Officer and military scientist Carl von Clausewitz defined two kinds of activity: "that of the *formation* and *conduct* of these single combats in themselves, and the *combination* of them with one another, with a view to the ultimate object of the War. The first is called *tactics,* the other *strategy.*"[14] Replace "combat" and "war" with "productive activity" and "business," and the same statement describes tactical and strategic business administration. Strategic management establishes corporate architecture, designs business models, anticipates market demands, plans product mixes, allocates resources to and coordinates efforts of various operations, evaluates performances, and adjusts plans,



all with long-term objectives, such as continuously maximizing return on investment. Tactical organization arranges finance, chooses and integrates manufacturing processes, coordinates workers and machines, arranges supply logistics, and decides on specific products, with criteria such as improving productivity. Operational organization aligns tactics to the strategy. Each depends crucially on relevant intelligence and is responsible for designing and maintaining efficient channels of information flow.

In the old days, when business firms were small and their operations simple, family management sufficed. As business scale and complexity escalated in the late nineteenth century, corporate organization became increasingly crucial. Strategic and tactical management were respectively pioneered by civil and mechanical engineers, who took on the new jobs to meet the developing challenges in their productive processes. Business historian Alfred Chandler wrote: "The most creative of those men who laid the groundwork for modern American corporate administration were three professionally trained engineers—Benjamin H. Latrobe, Daniel C. McCallum, and J. Edgar Thomson. Of the many contenders for the title of the founder of modern business administration, these three have the strongest claim."[15] All three were civil engineers who built railroads and later took to managing the operations of those railways.

Railroads, the first enterprise of sufficient scale operating at sufficiently high speed, brought a new kind of business that befuddled old-fashioned management. By the 1850s, the large rail lines were operating hundreds of locomotives and thousands of freight cars that carried loads millions of miles per year. Prompt and efficient operation of such complex systems, with numerous operators and agents and a haphazard flow of money, demanded a novel administrative approach. Civil engineers naturally filled the job, not only because they were familiar with the physical technology but also because they were already experts in planning, organization, and logistics.

Working independently on three railroad lines, Latrobe, McCallum, and Thomson applied their rational and analytic minds, honed in engineering practice, to solve the new managerial problems. They created the first elaborate internal organizational structures appropriate for private enterprise. Their organizations included separate central office

and regional divisions, distinct functional departments, clearly defined loci of authority and responsibility, and efficient channels of information flow. For the first time, information gathered within a corporation was used extensively to aid coordination and planning. With these innovations and "general principles of administration," they developed groundbreaking models for the structure of modern large corporations. Through the intermediate influence of financiers, their management methods became models for other businesses.

With the advent of mass production in the early twentieth century, giant manufacturing corporations appeared. Manufacturing was more dynamic than railroad operation and required more sophisticated administration. Prominent among those who completed the managerial revolution started by railroad engineers was a new generation of engineers who established the basic structures of American corporate capitalism: Sloan; Swope, who diversified the product lines of GE; Alfred, Coleman, Irenée, and Pierre du Pont, who reorganized a sleepy family company and turned it into a galloping giant. They were college-educated engineers in various fields, sometimes unrelated to the business of the corporations they came to head. Sloan was educated as an electrical engineer, although he spent his career in the automobile industry. What made them innovative executives was not specific pieces of engineering knowledge but their general technical aptitude and scientific approach to handling complex situations. Sloan wrote: "General Motors is an engineering organization . . . Many of our leading executives, myself among them, have an engineering background."[16] Applying the engineer's practicality, rationality, thoroughness, and scientific approach to financial control and corporate governance, those engineers designed a corporate structure at GM that became the model for large multidivisional corporations. "Reengineering corporations," a term popular in the last decade of the twentieth century, is apt because corporate structures were engineered in the century's early decades.